The longer-term impacts of Western diet on human cognition and the brain.

Animal work over the last three decades has generated a convincing body of evidence that a Western diet - one high in saturated fat and refined carbohydrates (HFS diet) - can damage various brain systems. In this review we examine whether there is evidence for this in humans, using converging lines of evidence from neuropsychological, epidemiological and neuroimaging data. Using the animal research as the organizing principal, we examined evidence for dietary induced impairments in frontal, limbic and hippocampal systems, and with their associated functions in learning, memory, cognition and hedonics. Evidence for the role of HFS diet in attention deficit disorder and in neurodegenerative conditions was also examined. While human research data is still at an early stage, there is evidence of an association between HFS diet and impaired cognitive function. Based upon the animal data, and a growing understanding of how HFS diets can disrupt brain function, we further suggest that there is a causal link running from HFS diet to impaired brain function in humans, and that HFS diets also contribute to the development of neurodegenerative conditions.